

Properties of Sum of Periodic Functions

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Introduction.

This journal entry explores different properties, constructions, and lemmata about the sum of periodic functions.

Specifically, these statements below, 4 statements and 4 constructions:

Lemma 1. Sum two continuous periodic functions with irrational ratio of periods is not periodic. (Pg. 1 to 2)

Constr. 1. Two periodic that sums to the identity function. (Pg. 2 to 3)

Constr. 2. Sum of two continuous periodic functions with irrational ratio of periods that **is** periodic (Pg. 4),

Lemma 2. Quadratic functions cannot be expressed as the sum of two periodic functions. (Pg. 4)

Constr. 3. Quadratic functions expressed as the sum of three periodic functions. (Pg. 5 to 6)

Lemma 3. n th order polynomial can not be expressed as the sum of n different periodic functions (generalized *Lemma 2*) (Pg. 6 to 7).

Constr. 4. n th order polynomial **can** be expressed as the sum of $n + 1$ different periodic functions. (generalized *Constr. 1* and 3) (Pg. 7 to 9)

Lemma 4. Exponential functions can not be expressed as the sum of any number of periodic functions. (Pg. 9)

All the constructions relies on the Axiom of Choice, or at least the Axiom of Choice on reals.

Keep in mind that all the constructions are also strictly discontinuous everywhere, strictly non-measurable under any normal measure, and almost impossible to be plotted or graphed.

Lemma 1

Defining forward difference:

$$\Delta(n)f(x) = f(x + n) - f(x)$$

This is going to be used as a notational shorthand.

Lemma. Given that two continuous periodic functions with irrational ratio of periods,

$$f(x + T_f) = f(x)$$

$$g(x + T_g) = g(x)$$

$$\frac{T_f}{T_g} \notin \mathbb{Q}$$

$h = f + g$ is not periodic, unless any of those function are constant functions.

Proof. We can proof this by contradiction, first assuming h is periodic, with period T_h

Proposition 1.

$$\Delta(T_f)h(x) = \Delta(T_f)g(x) \text{ is constant}$$

Proof. Observe that since $f(x + T_f) = f(x)$, and h consists of only f and g , we can have the following observation.

Observation.

$$h(x + T_f) - h(x) = g(x + T_f) - g(x + T_f)$$

$$\Delta(T_f)h(x) = \Delta(T_f)g(x)$$

Now observe some properties about this statement.

- RHS(Right Hand Side) have period T_g , since x is only shifted then inputted into g which have period T_g ,
- LHS(Left Hand Side) have period T_h with the same logic.

- Both sides are continuous, since it consists only of sum of shifted and scaled continuous function f, g .

Since there is no continuous function with two distinct unique period. From these three properties, we can deduce that

$$\Delta(T_f)h(x) / \Delta(T_f)g(x)$$

is constant regardless of x .

■

Proposition 2.

$$\Delta(T_f)h(x) = \Delta(T_f)g(x) = 0$$

Proof.

$$\begin{aligned} & \int_0^{T_h} \Delta(T_f)h(x) \, dx \\ &= \int_0^{T_h} h(x + T_f) \, dx - \int_0^{T_h} h(x) \, dx \\ &= \int_{T_f}^{T_h+T_f} h(u) \, du - \int_0^{T_h} h(x) \, dx \end{aligned}$$

Now, using the periodicity of h , we can get the following equality

$$\int_a^{T_h+a} h(u) \, du = \int_0^{T_h} h(u) \, du$$

Which means the two sides of the previous minus side is strictly equal.

Therefore

$$\begin{aligned} & \int_0^{T_h} \Delta(T_f)h(x) \, dx \\ &= \int_0^{T_h} h(x) \, dx - \int_0^{T_h} h(x) \, dx \\ &= 0 \end{aligned}$$

■

Therefore, since continuity implies Riemann integrability, T_f must also be a period of $g(x)$. $g(x)$ is continuous, with two different periods, must be a constant function too.

Q.E.D.

Construction 1

Lemma. Exist periodic function f and g , such that $f(x) + g(x) = x$, assuming axiom of choice, or axiom of choice on the reals.

Proof. See below construction.

Construct f and g have to be period with irrational ratio. Here let's make two arbitrary choice. 1 and π

Making another completely arbitrary choice $f(0) = g(0) = 0$, since shifting functions up and down does not affect the periodicity.

From these two assumptions we can have the following chain of reasoning.

1.

$$\begin{aligned} f(n) &= 0 \\ g(n\pi) &= 0 \end{aligned} \quad \text{(Using periodicity)}$$

2.

$$\begin{aligned} f(n\pi) &= n\pi \\ g(n) &= n \end{aligned} \quad \text{(Since we need } f(x) + g(x) = x, \text{ and one side is garenteed 0)}$$

3.

$$\begin{aligned} f(n + m\pi) &= m\pi \\ g(n + m\pi) &= n \end{aligned} \quad \text{(Using periodicity, again)}$$

(It satisfy our equation! Yay!)

The above is a valid construction for all of $\{n + m\pi : n, m \in \mathbb{Z}\}$.

To extend this function to the set of reals, we can first make the following observation.

Corollary. $\mathbb{R}$ can be split into **non-intersecting** subsets of shifted versions of $\{n + m\pi : n, m \in \mathbb{Z}\}$

A more constructive way to do this is just to use equivalence classes.

Define $a \sim b$ to be equivalent to the statement:

exists $n, m \in \mathbb{N}$, such that $a - b = n + m\pi$

This satisfy all three properties about equivalence relationships:

1. $a - a = 0 + 0\pi$
 $\therefore a \sim a$ for all $a \in \mathbb{R}$ (reflectivity)
2. $a - b = n_1 + m_1\pi$ and $b - c = n_2 + m_2\pi$
 $\Rightarrow a - c = (n_1 + n_2) + (m_1 + m_2)\pi$
 $\therefore a \sim b$ and $b \sim c \Rightarrow a \sim c$ (transitivity)
3. $a - b = n + m\pi \Rightarrow b - a = (-n) + (-m)\pi$
 $\therefore a \sim b \Rightarrow b \sim a$ (symmetry)

As described by the *Fundamental Theorem of Equivalence Relations*, equivalence classes corresponds to partitions.

$$[\alpha] = \{\alpha + n + m\pi : n, m \in \mathbb{Z}\}$$

With the axiom of choice, we can choose one element from every one of the partitions / equivalence classes $[\alpha]$, now using it for the pivot

$$\begin{aligned} f(\alpha) &= g(\alpha) = \frac{\alpha}{2} \\ f(\alpha) + g(\alpha) &= \alpha \end{aligned}$$

(This choice is again completely arbitrary, you can choose any $a + b = \alpha$)

and run the same logic as before.

1.

$$\begin{aligned} f(n + \alpha) &= \frac{\alpha}{2} \\ g(n\pi + \alpha) &= \frac{\alpha}{2} \end{aligned}$$

(Using periodicity)

2.

$$\begin{aligned} f(n\pi + \alpha) &= n\pi + \frac{\alpha}{2} \\ g(n) &= n + \frac{\alpha}{2} \end{aligned}$$

(Since we need $f(x) + g(x) = x$, and one side is guaranteed $\frac{\alpha}{2}$)

3.

$$\begin{aligned} f(n + m\pi + \alpha) &= m\pi + \frac{\alpha}{2} \\ g(n + m\pi + \alpha) &= n + \frac{\alpha}{2} \end{aligned}$$

(Using periodicity, again)

(It again satisfy our equation! Yay!)

The above is now a valid construction for all of $\{\alpha + n + m\pi : n, m \in \mathbb{Z}\}$.

Using this construction for all of the partitions we already got, finding the union, we have successfully covered the entire number line.

Therefore, solved the functional equation for the entire number line.

One way of explicitly writing the function out could look like this:

$$\begin{aligned} f(x) &= \begin{cases} b\pi & \text{if } \dots, x = a + b\pi \\ b\pi + \frac{\sqrt{2}}{2} & \text{if } \dots, x = a + b\pi + \sqrt{2} \\ b\pi + \frac{\varphi}{2} & \text{if } \dots, x = a + b\pi + \varphi \\ \dots & \end{cases} \\ g(x) &= \begin{cases} a & \text{if } \dots, x = a + b\pi \\ a + \frac{\sqrt{2}}{2} & \text{if } \dots, x = a + b\pi + \sqrt{2} \\ a + \frac{\varphi}{2} & \text{if } \dots, x = a + b\pi + \varphi \\ \dots & \end{cases} \end{aligned}$$

(Omitting the quantifier, $\exists a, b \in \mathbb{Z}$,
 $\exists$ meaning exists)

$$\begin{aligned} f(x + 1) &= f(x) \\ g(x + \pi) &= g(x) \end{aligned}$$

Q.E.D.

Construction 2

Lemma. Exists two continuous periodic functions with irrational ratio of periods that sums to a periodic function.

Proof. We can reuse the previously constructed equivalence relationship.

$$x = n + m\pi + \alpha,$$
$$[x] = \{n + m\pi + \alpha : n, m \in \mathbb{Z}\}$$

Now, again, uses the axiom of choice to uniquely choose one possible pivot of α for each of the equivalence relationship. Making all real numbers have an unique representation in form

$$x = n + m\pi + \alpha : n, m \in \mathbb{Z}$$

With this representation, we can easily construct f and g to satisfy our condition:

$$f(x) = f(n + m\pi + \alpha) = n$$
$$g(x) = g(n + m\pi + \alpha) = -m$$

Observe that f have period π , whilst g have period 1.

$$f(x + \pi) = f(n + (m + 1)\pi + \alpha) = n$$
$$g(x + 1) = g((n + 1) + m\pi + \alpha) = -m$$

(Observe that the RHS is the same as above)

(We have the same pivot α as all those numbers are in the same equivalence class)

(Uniquely chosen previously)

At the same time, their sum have period $1 + \pi$

$$f(x + 1 + \pi) + g(x + 1 + \pi)$$
$$= f((n + 1) + (m + 1)\pi + \alpha)$$
$$+ g((n + 1) + (m + 1)\pi + \alpha)$$
$$= (n + 1) - (m + 1)$$
$$= n - m$$
$$= f(x) + g(x)$$

Satisfying our Lemma.

Q.E.D.

Lemma 2

Lemma. The function $f(x) = x^2$ cannot possibly be expressed as the sum of *two* periodic functions.

Proof. Proof by contradiction.

Assuming it is possible to expressed $f(x) = x^2$ as periodic $g(x) + h(x)$, with period a and b , then we can construct

$$f(x + a) - f(x)$$
$$= a^2 + 2ax$$
$$= g(x + a) - g(a) + f(x + a) - f(a)$$

Notice that $g(x + a) - g(a) = 0$, from the periodicity.

Which gets us to

$$f(x + a) - f(a) = a^2 + 2ax$$

LHS have have period b , RHS is a linear function. They can not possible be equal and this statement can not possible be true.

We formed a contradiction, so therefore the original statement is false.

Q.E.D.

Construction 3

Lemma. Exist periodic function f and g, h , such that $f(x) + g(x) + h(x) = x^2$, again, assuming axiom of choice, or axiom of choice on the reals.

Proof. We start with constructing a new equivalence class, similar to the previous one. But this time, instead with three different arbitrary linear-independant numbers.

Define $a \sim b$ to be equivalent to the statement:

$$\begin{aligned} &\text{exists } n, m, o \in \mathbb{N}, \\ &\text{such that } a - b = n + m\pi + o\varphi \end{aligned}$$

This equivalence class splits the set of reals into disjointed shifted copies of

$$\{n + m\pi + o\varphi : n, m, o \in \mathbb{N}\}$$

and each real belongs to exactly one of the equivalence classes

$$[x] = \{n + m\pi + o\varphi + \alpha : n, m, o \in \mathbb{N}\}$$

Again applying the axiom of choice, we can again make all real numbers have an **unique** representation in the form

$$x = n + m\pi + o\varphi + \alpha : n, m, o \in \mathbb{N}$$

For our final target sum, (x^2) , we can express it by squaring our representation.

$$\begin{aligned} x^2 &= (n + m\pi + o\varphi + \alpha)^2 \\ x^2 &= n^2 + nm\pi + no\varphi + n\alpha \\ &\quad + m\pi n + m^2\pi^2 + m\pi o\varphi + m\pi\alpha \\ &\quad + o\varphi n + o\varphi m\pi + o^2\varphi^2 + o\varphi\alpha \\ &\quad + \alpha n + \alpha m\pi + \alpha o\varphi + \alpha^2 \\ &= n^2 \\ &\quad + 2nm\pi + m^2\pi^2 \\ &\quad + 2no\varphi + 2mo\varphi\pi + o^2\varphi^2 \\ &\quad + 2\alpha n + 2\alpha m\pi + 2\alpha o\varphi + \alpha^2 \end{aligned}$$

Here, the path diverge a bit from the previous proof. Observe that each of the terms in our target result actually have their own periodicity.

We can categorize them into the below categories:

We can treat α as the constant term when we stay in the same equivalence class. Therefore it have three different periods, $1, \pi, \varphi$

$$\alpha^2 \text{ remain same when } x \Rightarrow x + n + m\pi + o\varphi$$

Each of the single variable term have two periods. (still, as previously stated, considering α as a constant)

$$n^2 \text{ and } \alpha n : x \Rightarrow x + m\pi + o\varphi$$

$$m^2 \text{ and } \alpha m : x \Rightarrow x + n + o\varphi$$

$$o^2 \text{ and } \alpha o : x \Rightarrow x + n + m\pi$$

On the other hand, every term relying on two variables have only one periods, which is the intersection. Since for each variable when the input is changed by any of the two periods, the value doesn't change, and none of their components will change when you take the intersection.

$$mo \text{ stays the same when } x \Rightarrow x + n$$

$$no \text{ stays the same when } x \Rightarrow x + m\pi$$

$$nm \text{ stays the same when } x \Rightarrow x + o\varphi$$

With these three observations, we can now quite easily construct our three functions, with our three different chosen periods, $1, \pi, \varphi$.

Start by putting all terms with just a single period into our designated function.

$$f(x) = f(n + m\pi + o\varphi + \alpha)$$

$$= 2mo\varphi\pi + \dots$$

$$g(x) = g(n + m\pi + o\varphi + \alpha)$$

$$= 2no\varphi + \dots$$

$$h(x) = h(n + m\pi + o\varphi + \alpha)$$

$$= 2nm\pi + \dots$$

For the terms with two different periods. We can put them into whatever function with the designated period we want, or even split them into two. A more regular example would be :

$$f(x) = f(n + m\pi + o\varphi + \alpha)$$

$$= 2mo\varphi\pi + m^2\pi^2 + 2\alpha m\pi$$

$$g(x) = g(n + m\pi + o\varphi + \alpha)$$

$$= 2no\varphi + o^2\varphi^2 + 2\alpha o\varphi$$

$$\begin{aligned} h(x) &= h(n + m\pi + o\varphi + \alpha) \\ &= 2nm\pi + n^2 + 2\alpha n \end{aligned}$$

Whilst a random irregular example that can still work as a solution could be:

$$\begin{aligned} f(n + m\pi + o\varphi + \alpha) \\ &= 2mo\varphi\pi + m^2\pi^2 - 2\alpha m\pi + (1 - \sqrt{2})o^2\varphi^2 \\ g(n + m\pi + o\varphi + \alpha) \\ &= 2no\varphi + \sqrt{2}o^2\varphi^2 + \alpha o + \frac{1}{3}n^2 \\ h(n + m\pi + o\varphi + \alpha) \\ &= 2nm\pi + \frac{2}{3}n^2 + 2\alpha n + 4\alpha m\pi \end{aligned}$$

(Yes this is a bit absurd,

but it still sums to the same thing as above,

as well as having the same periodicity)

Lastly, we can put our last “constant” term anywhere we want:

$$\begin{aligned} f(x) &= f(n + m\pi + o\varphi + \alpha) \\ &= 2mo\varphi\pi + m^2\pi^2 + 2\alpha m\pi + \frac{1}{3}\alpha^3 \\ g(x) &= g(n + m\pi + o\varphi + \alpha) \\ &= 2no\varphi + o^2\varphi^2 + 2\alpha o\varphi + \frac{1}{3}\alpha^3 \\ h(x) &= h(n + m\pi + o\varphi + \alpha) \\ &= 2nm\pi + n^2 + 2\alpha n + \frac{1}{3}\alpha^3 \end{aligned}$$

Here I’m splitting it equally to make my result symmetric, and with this, we have successfully constructed three periodic functions, that sums to the cubic function.

Q.E.D.

Lemma 3

Lemma. Any n th degree polynomial cannot be expressed as the sum of n periodic functions (or any lower).

Proof. First we need to introduce an observation:

Observation. Forward difference reduces the order of the polynomial by 1.

Proof. For any terms,

$$(x + h)^n - x^n$$

is the forward difference. LHS can be factorized with the binomial theorem:

$$\begin{aligned} (x + h)^n \\ &= \sum_{k=0}^n \binom{n}{k} x^{n-k} h^k \\ &= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} h + \binom{n}{2} x^{n-2} h^2 + \dots \end{aligned}$$

Observe that since $\binom{n}{0} = 1$, the highest term is always $1x^n$, and when we reduce by x^n , the highest term always get cancelled out. Therefore the result of the forward difference can not possibly have the highest term, therefore having 1 less order. ■

With this observation, we can proof our statement by contradiction.

Assuming it is possible to expressed polynomial P as the sum of n periodic functions $f_1, f_2, f_3, \dots, f_n$ and period $p_1, p_2, p_3, \dots, p_n$

$$P(x) = \sum_{i=1}^n f_i(x)$$

We can choose one of the period like p_m , then construct

$$P(x + p_m) - P(x) = \sum_{i=1}^n f_i(x + p_m) - f_i(x)$$

Observe that in this situation, LHS is an $n - 1$ th order polynomial, whilst on the RHS is $n - 1$ periodic functions, since f_m have been cancelled.

$$f_m(x + p_m) - f_m(x) = 0$$

Repeating this procedure, we can obtain with the result that a quadratic can be expressed by two periodic function (previously disproved in *Lemma. 2*), and even one linear function can be expressed by one periodic function (yeah no definitely not)

We formed a contradiction, so therefore we have disproved our original statement.

Q.E.D.

Construction 4

Lemma. For every polynomial with degree n , there exist $n + 1$ many periodic function that sums to it, again, assuming axiom of choice, or axiom of choice on the reals.

Proof. We first start by finding $n + 1$ many co-algebraic independant numbers.

We, again, start with constructing a new equivalence class, similar to the previous one. But this time, instead with $n + 1$ different arbitrary co-algebraic-independant numbers, meaning that they are far apart enough. For example $1, \pi, e, \varphi$.

To generate infinite many of them systematically explicitly in our construction, we can taking the logarithm of primes.

$$\begin{aligned} &\{\ln(p_n) : n \in \mathbb{N}\} \\ &\{\ln 2, \ln 3, \ln 5, \ln 7, \ln 11, \dots\} \end{aligned}$$

Defining new $a \sim b$ to be equivalent to the statement:

$$\begin{aligned} &\text{exists } l_1, l_2, \dots, l_{n+1} \subset \mathbb{N}, \\ &\text{such that } a - b = \sum_{x=1}^{n+1} l_x \ln p_n \\ &= l_1 \ln 2 + l_2 \ln 3 + \dots \end{aligned}$$

This equivalence class splits the set of reals into disjointed shifted copies of

$$\{l_1 \ln 2 + l_2 \ln 3 + \dots : l_1, l_2, \dots, l_{n+1} \in \mathbb{N}\}$$

and each real belongs to exactly one of the equivalence classes

$$[x] = \{\alpha + l_1 \ln 2 + \dots : l_1, l_2, \dots, l_{n+1} \in \mathbb{N}\}$$

Again applying the axiom of choice, we can again make all real numbers have an **unique** representation in the form

$$x = \alpha + l_1 \ln 2 + l_2 \ln 3 + \dots + l_{n+1} \ln p_n,$$

with α again being the constant term.

The construction follow the same logic as before, we are going to factorize our x^n , using the multinomial theorem, a generalized version of the

binomial theorem, of which takes the following form:

$$(x_1 + \dots + x_t)^n = \sum \frac{n!}{n_1! \dots n_t!} x_1^{n_1} \dots x_t^{n_t}$$

$$n_1 + n_2 + \dots + n_t = n; 0 \leq n_i \leq n$$

Rephrasing it, this means that

$$(x_1 + \dots + x_t)^n$$

factorizes into all the terms such

$$\frac{n!}{n_1! \dots n_t!} x_1^{n_1} \dots x_t^{n_t}$$

with positive n_i that sums to n .

For example for our previous construction for x^2 , which is $n = 2$ and $t = 4$. We get

$$(x_1 + x_2 + x_3 + x_4)^2$$

$$= \sum \frac{2}{n_1! n_2! n_3! n_4!} x_1^{n_1} x_2^{n_2} x_3^{n_3} x_4^{n_4}$$

$$n_1 + n_2 + n_3 + n_4 = 2; 0 \leq n_i \leq n$$

Which contains the terms:

$$\frac{2}{2} x_1^2$$

$$+ \frac{2}{1} x_1 x_2 + \frac{2}{2} x_2^2$$

$$+ \frac{2}{1} x_1 x_3 + \frac{2}{1} x_2 x_3 + \frac{2}{2} x_3^2$$

$$+ \frac{2}{1} x_1 x_4 + \frac{2}{1} x_2 x_4 + \frac{2}{1} x_3 x_4 + \frac{2}{2} x_4^2$$

$$= x_1^2$$

$$+ 2x_1 x_2 + x_2^2$$

$$+ 2x_1 x_3 + 2x_2 x_3 + x_3^2$$

$$+ 2x_1 x_4 + 2x_2 x_4 + 2x_3 x_4 + x_4^2$$

Perfectly resembling our previous factorization in *Constr. 3*.

For our factorization here, we are factorizing the sum of $n + 2$ different variables to the power of n . The variables include the constant,

$$\alpha,$$

and $n + 1$ different independant numbers

$$l_1 \ln 2, l_2 \ln 3, l_3 \ln 5, \dots, l_{n+1} \ln p_{n+1}$$

Putting that into the formula, this means that we will have all terms with form

$$\frac{n!}{n_1! \dots n_{n+2}!}$$

$$\times \alpha^{n_1} (l_1 \ln 2)^{n_2} (l_2 \ln 3)^{n_3} \dots (l_{n+1} \ln p_{n+1})^{n_{n+2}}$$

Therefore, now, the question is equivalent to categorize every single one of these terms into $n + 1$ different periodic functions, similar to before.

So why is this always possible?

Let's closely analyze the periodicity of each term.

First, obviously, that the coefficient, containing a bunch of factorials does not change the periodicity of the term. So we can happily strip that away.

$$\alpha^{n_1} (l_1 \ln 2)^{n_2} (l_2 \ln 3)^{n_3} \dots (l_{n+1} \ln p_{n+1})^{n_{n+2}}$$

Now, again, using the same argument as before, since α is constant within the entire equivalence class, we can see that it will not remove any possible periods from the later terms, we can also happily strip that away too. This turns our term into

$$(l_1 \ln 2)^{n_2} (l_2 \ln 3)^{n_3} \dots (l_{n+1} \ln p_{n+1})^{n_{n+2}},$$

with

$$n_2 + n_3 + \dots + n_{n+2} \leq n; 0 \leq n_i \leq n$$

Since our original statement is equivalent to

$$n_2 + n_3 + \dots + n_{n+2} = n - n_1; 0 \leq n_i \leq n,$$

and now the non-negative n_1 is no longer required.

Now, away from all the distractions, we can analyze the periodicity of each term easily.

Observe that

$$l_m \ln p_m$$

have periodicity of all of the $\ln p_o$ except for $o = m$, for the simple reason that any increase of the variable x by any other of the $\ln p_o$ will lead to x being in the same equivalence class and some other l_o being increased by 1. This fact should be quite apparent to you if you have been reading from the start.

While at the same time, taking the exponent of a term doesn't change the periodicity (as long as it's not 0)

Therefore we can see that each of the term $(l_m \ln p_m)^n$ removes $\ln p_m$ from the period when $n \neq 0$, as obviously

$$\begin{aligned} \text{when } x &\Rightarrow x + \ln p_m, \\ (l_m \ln p_m)^n &\Rightarrow ((l_m + 1) \ln p_m)^n \end{aligned}$$

Which means therefore

$$(l_1 \ln 2)^{n_2} (l_2 \ln 3)^{n_3} \dots (l_{n+1} \ln p_{n+1})^{n_{n+2}},$$

have the periodicity including all the $\ln p_m$, except for the all $n_i > 0$.

But also remember that

$$n_2 + n_3 + \dots + n_{n+2} \leq n; \quad 0 \leq n_i \leq n.$$

Together with this, we are finally at our final step. Observe that left hand side is $n + 1$ integers, whilst the right side says the sum can at most be n . With the pigeonhole principle, we can see this means that the LHS have at one 0, which means that there is at least one periodicity in form $\ln p_n$ that haven't been excluded.

This means every term after the factorization have at least one period in form $\ln p_n$, of which we only have $n + 1$ many of.

We can put every term into the periodic function with the corresponding period of $\ln p_n$, of which, again, we only have $n + 1$ number of.

Therefore, any factorizations for any polynomial with order n , can be split into $n + 1$ many periodic functions!

Q.E.D.

Lemma 4.

Lemma. e^x can not be expressed as the sum of *any finite number* of periodic functions.

Proof. Prove by contradiction.

Assuming that it is possible to express e^x as the sum of at least n of periodic functions.

Then we can choose one of the period functions and cancel it out with

$$\begin{aligned} e^{x+p} - e^x \\ \text{(with } p \text{ being the period)} \end{aligned}$$

Supposedly this new function can be expressed as $n - 1$ many periodic functions, but at the same time:

$$\begin{aligned} e^{x+p} - e^x &= e^x + e^p - e^x \\ &= (e^p - 1)e^x \end{aligned}$$

That same expression is a multiple of e^x . This means e^x can be expressed by $n - 1$ many periodic functions, contradicting with our original statement. Therefore, it must be false.

Q.E.D.